

PIZZA SALES ANALYSIS

Project Overview:

Analyzed pizza sales data using SQL to identify sales trends, revenue, and top-performing pizzas. Created business insights using aggregations, joins, CTEs, and window functions.

01-Retrieve the total number of orders placed.

```
select count(order_id) as Total_orders
from orders;
```

	Total_orders
▶	21350

02-Calculate the total revenue generated from pizza sales.

```
select round(sum(order_details.quantity * pizzas.price),2) as total_sales
from order_details join pizzas
on pizzas.pizza_id = order_details.pizza_id;
```

Result Grid	
	total_sales
▶	817860.05

03-Identify the highest-priced pizza.

```
select pizza_types.name,pizzas.price from
pizza_types join pizzas on
pizzas.pizza_type_id = pizza_types.pizza_type_id
order by price desc
limit 3;
```

	name	price
▶	The Greek Pizza	35.95
	The Greek Pizza	25.5
	The Brie Carre Pizza	23.65

04-Identify the most common pizza size ordered.

```
select pizzas.size,count(order_details.order_details_id) as order_count
from pizzas join order_details
on pizzas.pizza_id = order_details.pizza_id
group by pizzas.size order by order_count desc;
```

	size	order_count
▶	L	18526
	M	15385
	S	14137
	XL	544
	XXL	28

05-List the top 5 most ordered pizza types along with their quantities

```
select pizza_types.name, sum(order_details.quantity) as quantity
from pizza_types
join pizzas on pizza_types.pizza_type_id = pizzas.pizza_type_id
join order_details on order_details.pizza_id = pizzas.pizza_id
group by pizza_types.name
order by quantity desc
limit 5;
```

	name	quantity
▶	The Classic Deluxe Pizza	2453
	The Barbecue Chicken Pizza	2432
	The Hawaiian Pizza	2422
	The Pepperoni Pizza	2418
	The Thai Chicken Pizza	2371

06-Join the necessary tables to find the total quantity of each pizza category ordered.

```
select pizza_types.category, sum(order_details.quantity) as quantity
from pizza_types
join pizzas on pizzas.pizza_type_id = pizza_types.pizza_type_id
join order_details on pizzas.pizza_id = order_details.pizza_id
group by pizza_types.category
order by quantity;
```

	category	quantity
▶	Chicken	11050
	Veggie	11649
	Supreme	11987
	Classic	14888

07-Determine the distribution of orders by hour of the day.

```
select HOUR(order_time) as hour, count(order_id) as order_count
from orders
group by hour(order_time);
```

	hour	order_count
▶	9	1
	10	8
	11	1231
	12	2520
	13	2455
	14	1472
	15	1468
	16	1920
	17	2336
	18	2399
	19	2009
	20	1642

08-Join relevant tables to find the category-wise distribution of pizzas.

```
select category,count(name) from pizza_types
group by category;
```

	category	count(name)
▶	Chicken	6
	Classic	8
	Supreme	9
	Veggie	9

09-Group the orders by date and calculate the average number of pizzas ordered per day.

```
SELECT
    AVG(total_pizzas) AS average_pizzas_per_day
FROM (
    SELECT
        orders.order_date,
        SUM(order_details.quantity) AS total_pizzas
    FROM orders
    JOIN order_details
        ON orders.order_id = order_details.order_id
    GROUP BY orders.order_date
) AS daily_sales;
```

	average_pizzas_per_day
▶	138.4749

10-Determine the top 3 most ordered pizza types based on revenue.

```
SELECT
    pizza_types.name,
    SUM(order_details.quantity * pizzas.price) AS revenue
FROM pizza_types
JOIN pizzas
    ON pizza_types.pizza_type_id = pizzas.pizza_type_id
JOIN order_details
    ON order_details.pizza_id = pizzas.pizza_id
GROUP BY pizza_types.name
ORDER BY revenue DESC
LIMIT 3;
```

	name	revenue
►	The Thai Chicken Pizza	43434.25
	The Barbecue Chicken Pizza	42768
	The California Chicken Pizza	41409.5

11-Calculate the percentage contribution of each pizza

type to total revenue.

```
SELECT
    pizza_types.name,
    ROUND(
        SUM(order_details.quantity * pizzas.price)
        /
        (
            SELECT SUM(order_details.quantity * pizzas.price)
            FROM order_details
            JOIN pizzas
            ON pizzas.pizza_id = order_details.pizza_id
        ) * 100,
        2
    ) AS revenue_percentage
FROM pizza_types
JOIN pizzas
    ON pizza_types.pizza_type_id = pizzas.pizza_type_id
JOIN order_details
    ON order_details.pizza_id = pizzas.pizza_id
GROUP BY pizza_types.name
ORDER BY revenue_percentage DESC;
```

	name	revenue_percentage
►	The Thai Chicken Pizza	5.31
	The Barbecue Chicken Pizza	5.23
	The California Chicken Pizza	5.06
	The Classic Deluxe Pizza	4.67
	The Spicy Italian Pizza	4.26
	The Southwest Chicken Pizza	4.24
	The Italian Supreme Pizza	4.09
	The Hawaiian Pizza	3.95
	The Four Cheese Pizza	3.95
	The Sicilian Pizza	3.78
	The Pepperoni Pizza	3.69
	The Greek Pizza	3.48
	The Mexicana Pizza	3.27
	The Five Cheese Pizza	3.19
	The Pepper Salami Pizza	3.12
	The Italian Capocollo Pizza	3.07
	The Vegetables + Vegetable...	2.98
	The Prosciutto and Arugula ...	2.96
	The Napolitana Pizza	2.95

12-- Analyse the cumulative revenue generated over time.

```
SELECT
    order_date,
    SUM(revenue) OVER(ORDER BY order_date) AS cumulative_revenue
FROM
(
    SELECT
        orders.order_date,
        SUM(order_details.quantity * pizzas.price) AS revenue
    FROM orders
    JOIN order_details
        ON orders.order_id = order_details.order_id
    JOIN pizzas
        ON order_details.pizza_id = pizzas.pizza_id
    GROUP BY orders.order_date
) AS daily_sales
ORDER BY order_date;
```

	order_date	cumulative_revenue
▶	2015-01-01	2713.8500000000004
	2015-01-02	5445.75
	2015-01-03	8108.15
	2015-01-04	9863.6
	2015-01-05	11929.55
	2015-01-06	14358.5
	2015-01-07	16560.7
	2015-01-08	19399.05
	2015-01-09	21526.4
	2015-01-10	23990.350000000002
	2015-01-11	25862.65
	2015-01-12	27781.7
	2015-01-13	29831.300000000003
	2015-01-14	32358.700000000004
	2015-01-15	34343.500000000001
	2015-01-16	36937.650000000001
	2015-01-17	39001.750000000001
	2015-01-18	40978.600000000006
	2015-01-19	43365.750000000001